Strike Three, You're Out?

A Mathematical Analysis of How an Automated Strike Zone Could Alter Major League Baseball Hitter Performance

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Contents

1 Mathematical Preliminaries

Definition 1. *T has* ***Student t-Distribution*** *with parameter* $\nu = 1, 2, 3, \ldots$ *if its probability density function is*

$$f(x) = \begin{cases} \dfrac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \dfrac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, & x > 0, \\ 0 & \textit{otherwise.} \end{cases}$$

where Γ *is Euler's Gamma Function.*

Let $X_1, X_2, \ldots, X_n$ be independent normally distributed random variables with common mean μ_X and common variance σ_X^2.

Let $Y_1, Y_2, \ldots, Y_m$ be independent normally distributed random variables with common mean μ_Y and common variance σ_Y^2.

Note that:

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n} \qquad \bar{Y} = \frac{Y_1 + \cdots + Y_m}{m}$$

$$S_x^2 = \sum_{i=1}^{n} \frac{(X_i - \bar{X})^2}{n-1} \qquad S_y^2 = \sum_{i=1}^{m} \frac{(Y_i - \bar{Y})^2}{m-1}$$

Definition 2. *Suppose* $\mu_X = \mu_Y$*. The random variable*

$$T^* = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}}$$

has the ***Smith-Satterthwaite-Welch Distribution*** *with*

$$\nu^* = \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m}\right)^2}{\frac{\left(\frac{s_x^2}{n}\right)^2}{n-1} + \frac{\left(\frac{s_y^2}{m}\right)^2}{m-1}}$$

degrees of freedom.

Notice that ν^* may not end up being an integer in this situation.

Theorem 1. *For all real numbers a and b, where* $a < b$*,* $Prob(a \leq T^* \leq b)$ *is approximately* $P(a \leq T \leq b)$ *where T has the Student t-Distribution with* ν *degrees of freedom and* ν *is the positive integer closest to* ν^* *[31].*

The application of this theorem is called the **Welch's t-test**. Welch's t-test is used whenever a test involving the Student t-Distribution cannot be used, because we cannot assume that $\sigma_X^2 = \sigma_Y^2$, and when we cannot use the Central Limit Theorem because our sample size is not large enough [27].

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Definition 3. *The random variable x has **Normal Distribution** with parameters μ and $\sigma > 0$ if its probability density function is:*

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad -\infty < x < \infty$$

One key property of the Normal Distribution used in this paper is that the graph of the data forms a bell-shaped curve [20].

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Theorem 2. *If we use the notation described before the definition of the Welch Distribution earlier, then the random variable Z defined by*

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_Y)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_Y^2}{m}}}$$

has the Normal Distribution with parameters $\mu = \mu_X - \mu_Y$ and $\sigma^2 = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$ [16].

The application of this theorem is called the **Two-Sample z-test**.

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Definition 4. *A random variable Z follows the **Standard Normal Distribution** if it has mean 0 and variance 1. We can write this as $Z \sim N(0,1)$.*

Definition 5. *Z_n **converges to Z in distribution** if:*

For all a,b, where $b > a$:

$$\lim_{n\to\infty} P(a < Z_n < b) = P(a < Z < b)$$

For notation, this can be written as $Z_n \xrightarrow{d} N(0,1)$.

Theorem 3. ***Central Limit Theorem***

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables with mean μ and variance σ^2. Define

$$Z_n = \frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}}.$$

Let $Z \sim N(0,1)$. As $n \to \infty$, the sequence Z_n converges in distribution to Z [29].

—

Definition 6. *A random variable X has **Poisson Distribution** with parameter $\lambda > 0$, if its probability function, $f(x) = p(x : \lambda)$ is*

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2 \dots$$

Theorem 4. *Let X be a random variable having Poisson Distribution, with parameter $\lambda > 0$. Then, λ is both the mean and the variance [28].*

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Definition 7. ***Panel data*** *consist of many observations on the same individuals over several time periods [23].*

In this specific study, the data are classified as panel data as they contain metrics (strikeout percentage, age, etc...) on batters over several years.

The data used in this study are considered to be **unbalanced panel data**, as not every hitter has data for all years the study takes into consideration (2019-2025) [19].

Figure 1 (seen below) gives an example of how panel data look in this context.

	Season	Name	Team	Level	Age	PA	BB.
10642-2022	2022	Carlos Perez	COL	AAA	31	522	0.10344828
10642-2024	2024	Carlos Perez	OAK	AAA	33	468	0.10470085
10642-2025	2025	Carlos Perez	CHC	AAA	34	449	0.10913140
10655-2019	2019	Rob Brantly	PHI	AAA	29	272	0.11764706
10655-2021	2021	Rob Brantly	NYY	AAA	31	264	0.07575757
10655-2022	2022	Rob Brantly	NYY	AAA	32	201	0.05472637
10700-2019	2019	Ryan LaMarre	ATL	AAA	30	455	0.08351648
10700-2021	2021	Ryan LaMarre	NYY	AAA	32	240	0.11250000
10711-2019	2019	Arismendy Alcantara	NYM	AAA	27	339	0.09439528
10711-2021	2021	Arismendy Alcantara	SFG	AAA	29	255	0.08235294

Figure 1: Sample of panel data used in the study

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Definition 8. *The sample correlation coefficient of the paired data* $(X_1, Y_1), (X_2, Y_2), \ldots,$ (X_i, Y_i) *is defined as*

$$R = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

$\mathbf{R^2}$ is used to show how much of the variation in the dependent variable can be explained by the independent variables in a regression model. **$\mathbf{R^2}$ is also called Overall $\mathbf{R^2}$** [26].

(Note: R^2 values can only range from 0 to 1, with a value of 0 indicating a very poor model fit and a value of 1 being a perfect model fit)

Definition 9. ***Multicollinearity*** *is present in a regression model when two variables used as predictors are highly correlated, that is, if* R *is greater than 0.75.*

When multicollinearity is present, the predictor variables have overlapping information, making the coefficient estimates unreliable and preventing the model from isolating each variable's true effect on the dependent variable [12]

Definition 10. *Let* $X_1, X_2, \ldots, X_p$ *represent independent variables in a regression model. The* ***Variance Inflation Factor*** *value for a variable,* X_j*, is*

$$VIF_j = \frac{1}{1 - R_j^2},$$

where R_j^2 *is the* R^2 *value determined from the regression model represented by the equation*

$$X_j = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \cdots + \alpha_{j-1}X_{j-1} + \alpha_{j+1}X_{j+1} + \cdots + \alpha_p X_p + \epsilon_j$$

where α_i *is the coefficient for each variable and* ϵ_j *is the error term.*

Variance Inflation Factor is a metric that can show the degree of multicollinearity that is present with respect to a predictor variable, X_j.

Note that a variable found to have a Variance Inflation Factor value greater than 10 is considered to be highly correlated with other predictor variables and should be appropriately addressed [12].

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Definition 11. *An **individual fixed effect** is a trait that doesn't change within a player over time, such as handedness.*

Definition 12. *A **Fixed Effects Model** for panel data is one of the form*

$$Y_{i,t} = \beta_1 X_{1,i,t} + \beta_2 X_{2,i,t} + \cdots + \beta_j X_{j,i,t} + \alpha_i + \epsilon_{i,t},$$

where $Y_{i,t}$ is the value of the dependent variable for an individual i at time t. $X_{j,i,t}$ for $j = 1, 2, \ldots, n$ represents the predictor variables in the regression model for an individual i at time t, with corresponding coefficients β_j for $j = 1, 2, \ldots, n$. α_i represents the control for individual fixed effects, variables that stay constant over time for each individual, and $\epsilon_{i,t}$ is the error term.

Definition 13. *An error term is **idiosyncratic** if it has zero conditional mean with all the independent variables in a regression model over all time periods, that is,*

$$E[\epsilon_{i,t} | X_{i,1}, X_{i,2}, \ldots, X_{i,T}] = 0$$

Note that an assumption that must be true for Fixed Effects Models is that the error term must be idiosyncratic [21].

This assumption implies that the error term and all independent variables are not correlated, so that we are able to identify how time-varying characteristics that are accounted for in the model affect the dependent variable [14].

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Definition 14. *An **indicator variable** in a regression model is one that can only take on a value of 0 or 1, indicating thereby that an individual belongs to one category in a certain time period.*

The regression analysis on the dependent variables strikeout percentage, walk rate, and swinging strike percentage was performed by **Fixed Effects Ordinary Least Squares (OLS)** estimation with the following regression equation

$$Y_{it} = D_1 abs2023_{i,t} + D_2 abs2024_{i,t} + D_3 abs2025_{i,t} + B_1 Age_{i,t} + B_2 PA_{i,t} + \alpha_i + \epsilon_{i,t}.$$

where $abs2023_{i,t}$ is 1, if player i, played at least one year without the Automated Ball and Strike System and in the year 2023. Otherwise, $abs2023_{i,t}$ is 0. A similar definition exists for $abs2024_{i,t}$ and $abs2025_{i,t}$.

- $Y_{i,t}$ is the dependent variable (strikeout percentage, walk rate, or swinging strike percentage) for an individual i at time t.

- $Age_{i,t}$ and $PA_{i,t}$ denote a player's age and plate appearances, respectively, at a certain time t with corresponding coefficients, B_1 and B_2.
- α_i represents the control for individual fixed effects.
- D_1, D_2, and D_3 represent the coefficients for the ABS indicator variables for 2023, 2024, and 2025, respectively.

When interpreting the coefficients of indicator variables in a regression model, it is important to realize it is a comparison to a group omitted from the regression. In this case, the reference group is before ABS, that is, 2019-2022 for every player.

For example, if we are studying a player's strikeout rate in the year 2024, we would see the abs2024 variable have a value of 1, while the other two indicator variables would be 0. The interpretation of D_2 would be relative to the player's performance before ABS implementation compared to his performance in 2024 [22].

Definition 15. ***Clustered standard errors*** *are a way to estimate the standard error of a regression coefficient when data can be grouped into* ***clusters****.*

In this case, a cluster would represent a certain player seen multiple times in the data set. We use clustered standard errors to account for the fact that observations in each cluster would have correlated error terms, which violates a key assumption of OLS models. Thus, we can have more robust inferences on the regression coefficients with this technique [17].

Within $\mathbf{R^2}$ measures the proportion of the variation in the dependent variable, within individuals over time, that is explained by a regression model, **after removing individual fixed effects** [25].

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Definition 16. *A* ***Fixed Effects Poisson Model*** *is a Fixed Effects Model*

$$Y_{i,t} = \beta_1 X_{1,i,t} + \cdots + \beta_j X_{j,i,t} + \alpha_i$$

in which the dependent random variable $Y_{i,t}$ has Poisson Distribution with parameter $\mu_{i,t}$, $i = 1, 2, \ldots, N$, $t = 1, 2, \ldots, T$.

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Theorem 5. *If $Y_{i.t}$ is a dependent variable in a Fixed Effects Poisson Model, as in the preceding definition, the conditional probability function of $Y_{i,t}$ given values of $\beta_1, \ldots, B_j$, $X_{1,i,t}, \ldots, X_{j,i,t}$, α_i is*

$$f = \frac{e^{-\mu_{i,t}} \mu_{i,t}^{y_{i,t}}}{y_{i,t}!}.$$

Also,

$$\mu_{i,t} = \mathbb{E}(Y_{i,t} | \beta_1, \ldots, \beta_j, X_{1,i,t}, \ldots X_{j,i,t}, \alpha_i).$$

When we have a Fixed Effects Poisson Model, we will use the regression equation

$$ln(\mu_{i,t}) = \beta_1 X_{1,i,t} + \cdots + \beta_j X_{j,i,t} + \alpha_i$$

Note: There is no error term due to the fact that the Poisson distribution already accounts for the error term in the randomness of the distribution [24].

Definition 17. *An **offset** is a term added to the linear combination of predictor variables.*

An offset is used to account for the fact that the dependent variable measured in counts will naturally increase with more plate appearances. Applying an offset fixes a certain variable to a coefficient of 1, therefore causing the variable to be interpreted as a rate.

In this case, we could easily see that having more plate appearances would increase the number of balls, strikes, or pitches in a season, warranting the use of the offset for the plate appearances variable [9].

With these definitions, we can then can see the specific regression equation used for my analysis on the number of balls, strikes, and pitches seen in a season is

$$ln(\mathbb{E}[Y_{i,t}]) = D_1 abs2023_{i,t} + D_2 abs2024_{i,t} + D_3 abs2025_{i,t} + B_1 Age_{i,t} + \alpha_i + ln(PA_{i,t})$$

where D_i for $i = 1, 2, 3$ are the coefficients for the respective indicator variables, $ln(PA_{i,t})$ represents an offset for plate appearances, and α_i represents the fixed effects.

Definition 18. *If $Y_1, \ldots, Y_n$ is an independent random sample from a population with Poisson Distribution such that*

$$Var(\bar{Y}) > \frac{\mathbb{E}(\bar{Y})}{n},$$

*then the sample is **over-dispersed**.*

The presence of over-dispersion violates the assumption that the Poisson Distribution has equal mean and variance. Over-dispersion can also cause quite small standard errors.

Over-dispersion is common in real-world count data (present in this study), however using clustered standard errors can take care of this issue [18].

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McFadden's Pseudo $\mathbf{R^2}$ measures how well the model designed improves the prediction on the dependent variable compared to a model containing no independent variables.

Note: McFadden's Pseudo R^2 ranges from 0 to 1 with higher values indicating a better fitting model (tends to be higher with fixed effects) [11]

Definition 19. *For any random variable $Y_{i,t}$, the **squared correlation** is*

$$R^2_{corr} = corr(Y_{i,t}, \hat{Y}_{i,t})^2$$

where Y_{it} is the actual value and $\hat{Y}_{it}$ is the predicted value from the model.

Squared Correlation compares the actual values and the predicted values from the model.

Note: Squared Correlation also ranges from 0 to 1 with higher values indicating a better fitting model (tends to be higher with fixed effects) [30].

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Definition 20. *A **Difference in Difference (DiD) Model** is one of the form*

$$Y_{i,t} = B_0 + B_1 G_i + B_2 T_t + B_3(G_i T_t) + \epsilon_{i,t}$$

where $Y_{i,t}$ is the dependent variable, B_0 is the intercept, G_i being an indicator variable that equals 1 if an individual is in the treatment group and 0 if in the control group, T_t being a similar indicator variable concerning time (where a value of 1 means post-treatment time period), and the term, $G_i T_t$, being the interaction term to measure how the treatment group changes relevant to the control group, and $\epsilon_{i,t}$ being the error term.

Note that β_1 gives the initial difference between the treatment and control group, β_2 does the same, only showing the change between the two different time periods relative to the control group, and β_3 shows the additional effect of being in the treatment group during the designated time period.

A Difference in Difference Model is used on panel data to observe trends over time, with a well-defined treatment and control group over the time studied.

A final note about Difference in Difference Models is that they have a "parallel trends" assumption, meaning that the both groups were on similar trajectories before the treatment was implemented [15].

2 Application and Statistical Tests

2.1 Introduction

A study by Boston University found that in 2018, umpires missed around 34,000 ball and strike calls throughout the season, an average of 14 per game. This same study found that there were 55 games that ended on an incorrect pitch call in the same season [7]. These are a few of the eye-opening statistics that show umpires may not have as keen sight as we thought. Umpire accuracy has been a hot topic of discussion throughout the baseball world for the past few years. Much of this discussion has been focused on trying to convince Major League Baseball (MLB) to implement an Automated Ball and Strike (ABS) system to call pitches and end any discrepancies that continue to exist when it comes to determining if a pitch is a ball or strike. The constant push for the ABS system obviously has been heard as Major League Baseball plans to implement an ABS Challenge System starting in the 2026 season [1]. With such a major juncture coming to the game of baseball, it is normal to wonder what may result from it. Will the integrity of the game be changed? How will player performance be affected? These are some of the questions I hope to bring insight to throughout my study. Before we jump any further, it is important to discuss how ABS works, was introduced, and why it is considered to be reliable.

2.2 How ABS Works

It is important to note that were two versions of ABS that were considered to be implemented into the MLB: Full ABS and the ABS Challenge System. The main difference between the two versions is how involved the umpire is in making decisions on whether a pitch is a ball or strike. In Full ABS, the umpire acts solely as a translator by reporting if the robot umpire called the pitch as a ball or a strike. There is no human judgment involved in the Full ABS version. Conversely, in the ABS Challenge System, the human umpire makes calls as was standard in the past, but with one caveat: each team has two challenges available if they question the call by the umpire. A challenge is considered successful whenever the umpire's original ruling was reversed. If a challenge is deemed successful, it results in a team retaining their challenge. In this version, we can receive the best of both worlds, keeping the human aspect to the game and implementing technology so there is an opportunity to ensure important calls are made correctly. This reason that I just described is one of many deciding factors on why the MLB chose the Challenge System to be the best fit for the game [2].

2.3 Hawk-Eye Technology and Strike Zone Construction

In the prior section, the concept of an robot umpire was discussed but not introduced. For anyone who has watched a game of baseball, it may be a bit troubling to figure out how a robot umpire and a human umpire can be present in the same space without any conflict. This is why Major League Baseball and its affiliates partnered with Hawk-Eye Innovations to utilize their high speed cameras to track a pitch's location and work in the background along with ABS. To ensure all dimensions of a pitch's location are identified, all ballparks are equipped with two cameras behind home plate, one along first base, one along third base, and one in center field. Utilizing these high speed cameras to design a three-dimensional strike zone allows the game to be more accurate and also doesn't eliminate the presence of a human umpire, who can approve challenge attempts and change calls that the technology definitely missed, a rare occurrence. Hawk-Eye technology, along with the ABS system doesn't just enhance the accuracy of the pitch calls, it also creates a strike zone area that is fair for batters [3]. Before ABS was introduced, the strike zone was considered to be more lenient to pitchers due to its rounded shape, with the strike zone being found as high as 55 percent of the batters height and as low as 24 percent of the batters height. MLB also found that the strike zone could be altered by umpires based on the pitch count. For example, it was found that in a 2-2 count (2 balls and 2 strikes) the human umpire strike zone was six square inches bigger compared to the ABS designed zone. This shows that there was obviously not a uniform strike zone set in place, which caused a huge disadvantage for batters. However, the ABS strike zone is rigidly defined as it takes the batter's height into account, which is measured at the beginning of the season to ensure uniformity. This way the strike zone does not penalize hitters in different situations and cannot benefit or limit hitters based on their height or batting stance [2]. See Figure 2 to depict the dimensions the ABS strike zone:

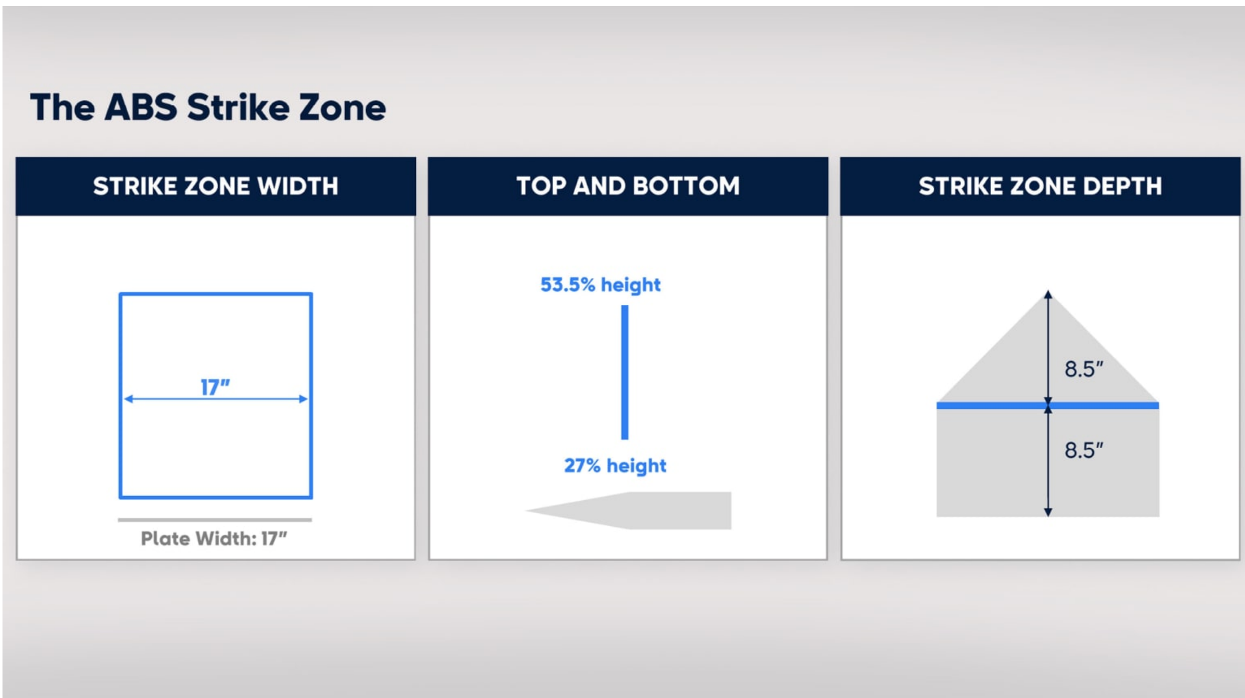


Figure 2: Dimensions of the ABS Strike Zone [2]

2.4 ABS Testing

All of this innovative technology being introduced into the game ultimately forced Major League Baseball's hand, prompting the league to begin testing these new systems in live-game settings. However, Major League Baseball did not want to immediately alter the highest level of the game with this new strike zone technology due to its unproven success. Therefore, Major League Baseball decided to implement the ABS system in Minor League Baseball (MiLB) first to see if it had a positive effect on the game. In 2019, the ABS system was first introduced in the independent Atlantic League, whose teams have no affiliation with a Major League Baseball team as Minor League teams do. The Atlantic League was the first baseball league to extensively see the use of the Full ABS system, as it was used from 2019-2022 in all games that were played. The first time that ABS was introduced into a league that is affiliated with MLB was in 2022 in the Florida State League, which is a Single-A League (lowest level of Minor League Baseball). During 2022, the Florida State League experimented with both the Full ABS system and the ABS challenge system. Taking both leagues into account, it was found that the full ABS system was effective, but increased the game duration and displeased fans and players alike by taking the human aspect out of the game. Having received this initial feedback, Major League Baseball decided to test both versions of the Automated Ball-Strike (ABS) system in Triple-A, the highest level of Minor League Baseball, to determine whether differences in skill level were responsible for the preference toward the ABS challenge system. Therefore, Major League Baseball and its Triple-A affiliates decided to provide a mix of both systems throughout the season. For some context, a Minor League Baseball team's schedule consists of weekly series against a certain team. It also important to note that Minor League teams usually play this weekly series from Tuesday to Sunday with one game played per day, barring any weather delays [6]. This scheduling quirk was the key to implementing the mix of the ABS systems as the Full ABS system was used for games on Tuesday-Thursday and the ABS challenge system was used for Friday-Sunday. This scheduling technique was used for the full 2023 season in Triple-A. This worked smoothly throughout 2023, so Triple-A began the 2024 season using this same format. In June 2024, Rob Manfred, the MLB commissioner, then announced that the ABS Challenge system would be used for all games for the remainder of the 2024 Triple-A season, as many fans and players felt it did not alter the the integrity of the game as much as the Full ABS system. After experimenting with what system the Minor League players and fans preferred, it was then time to begin implementing the ABS Challenge system into some MLB settings. Therefore, MLB decided to use the ABS Challenge system throughout Spring Training, a month of preseason games, and the All-Star Game in 2025. The impact of ABS was welcomed by the majority of Major League players, and this success is what led to Manfred announcing that the ABS Challenge system will be used throughout the 2026 MLB season [1].

Knowing that the ABS Challenge System is coming to Major League Baseball should not raise significant concern that the sport will lose its human element. However, an important question that arises is how player performance may be affected by this new aspect of the game. Unsurprisingly, several studies have already attempted to examine how various components of baseball have changed following the introduction of ABS. For instance, a notable study investigated how the ABS Challenge System influenced player performance in the Korean Baseball Organization (KBO). In 2024, the KBO became the first professional baseball league worldwide to implement an ABS system, allowing researchers to analyze its impact on player performance throughout the season. Using data from before and during the implementation, the authors examined how different pitch types were affected by the more rigid strike zone and whether batters became more aggressive in certain areas of the zone [8]. Upon reviewing this study and others like it, I noticed that most existing research approaches the topic from the pitcher's perspective. This pattern motivated me to conduct a study focusing instead on hitters, aiming to analyze potential changes in offensive performance once ABS is introduced in Major League Baseball. Specifically, I plan to evaluate how ABS may influence key hitting metrics such as strikeout rate, walk rate, swinging strike percentage, and the number of balls, strikes, and pitches seen throughout a season.

2.5 Data Collection

To evaluate how the Automated Ball-Strike system may influence a batter's strikeout rate, walk rate, swinging strike rate, and overall pitch outcomes, this study begins by identifying where the ABS system has previously been implemented. By focusing on Triple-A baseball, a league that has tested the ABS system for multiple years, I collected relevant batting data from this environment to compare player performance in both automated and traditional umpiring conditions. Some of the relevant variables that I collected for each batter were :

- Strikeout Rate (K%): Proportion of plate appearances that resulted in a strikeout.

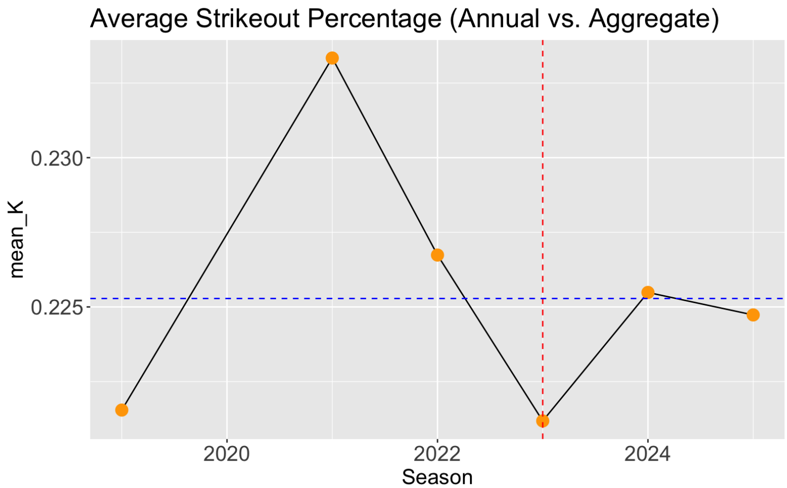
- Walk Rate (BB%): Proportion of plate appearances that resulted in a walk.
- Swinging Strike Rate (SwStr%): Proportion of pitches that a batter swings and misses at.
- Balls: Total number of balls a batter sees throughout a season.
- Strikes: Total number of called strikes a batter sees throughout a season.
- Pitches: Total number of pitches a batter sees throughout a season.
- Age
- Season
- Plate Appearances (PA)
- Team

Data for this study were obtained from FanGraphs [4], covering the seasons from 2019 through 2025. The sample includes all Triple-A batters who recorded at least 200 plate appearances in a given season. After filtering based on this criteria, the dataset contains 2,068 total player-season observations, representing 1,135 distinct players over the study period. One notable exception is the year 2020, for which no data are available due to the cancellation of the Triple-A season as a result of the COVID-19 pandemic.

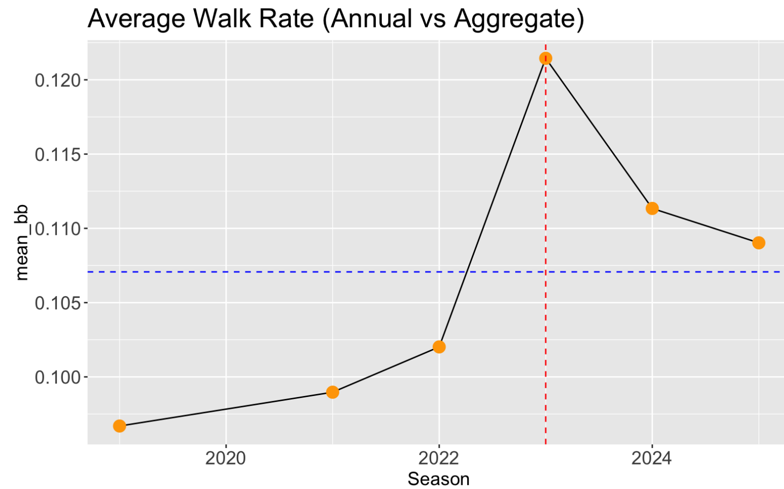
Due to data limitations, this study treats the Full ABS and ABS Challenge systems as equivalent. Although these systems differ in implementation, combining them strengthens the robustness of the analysis by maintaining sufficient sample sizes, since each system was only used in approximately half of Triple-A games during the 2023 and 2024 seasons in which the ABS was tested.

2.6 Initial Observations of Data

After examining the dataset's features, I generated time-series graphs for each performance metric to observe how league averages evolved between 2019 and 2025. Each graph includes a dotted blue line representing the mean of that metric compiled across all seasons, allowing for quick comparison of whether a given year's average, represented by the orange points, was above or below the long-term trend. Additionally, a red vertical line marks 2023, the year the Automated Ball-Strike (ABS) system was first implemented full-time in Triple-A Baseball, to help visualize any noticeable shifts in the data before and after its introduction.

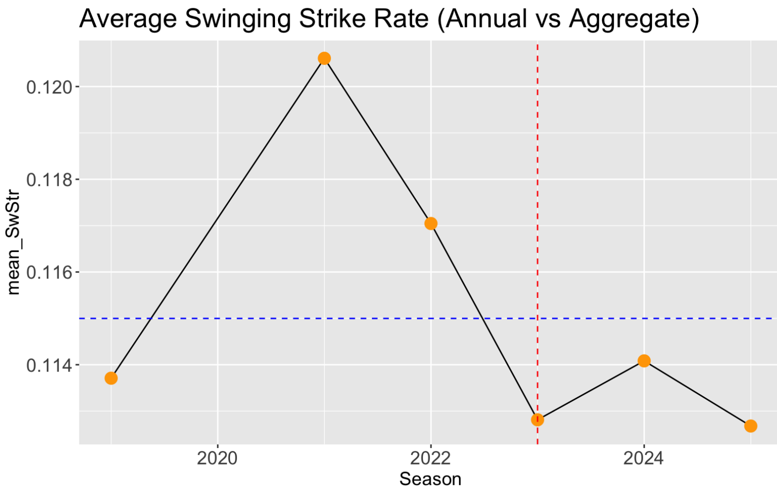


(a) **Strikeout Percentage (K%)**

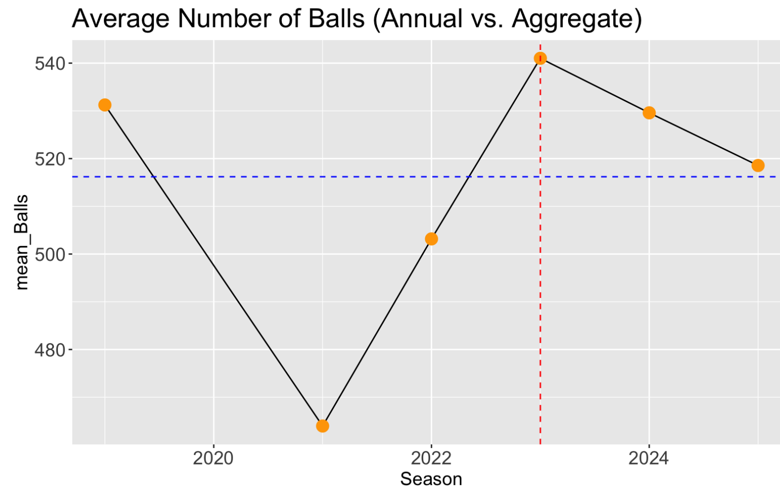


(b) **Walk Rate (BB%)**

Figure 3: **Comparisons of annual vs aggregate statistics for Strikeout Rate and Walk Rate.**

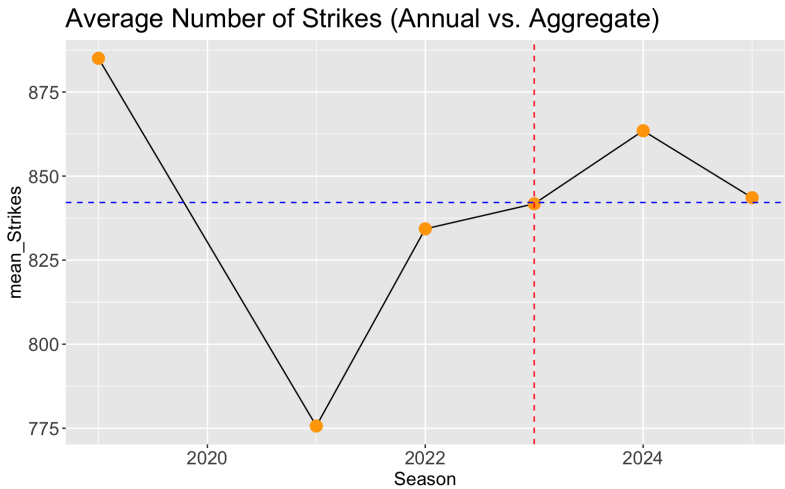


(a) **Swinging Strike Rate (SwStr%)**

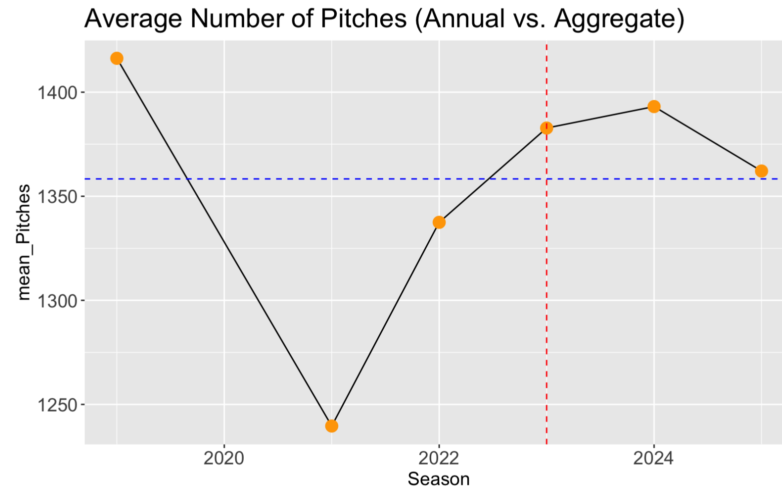


(b) **Balls**

Figure 4: **Comparisons of annual vs aggregate statistics for Swinging Strike Rate and Balls.**



(a) **Strikes**



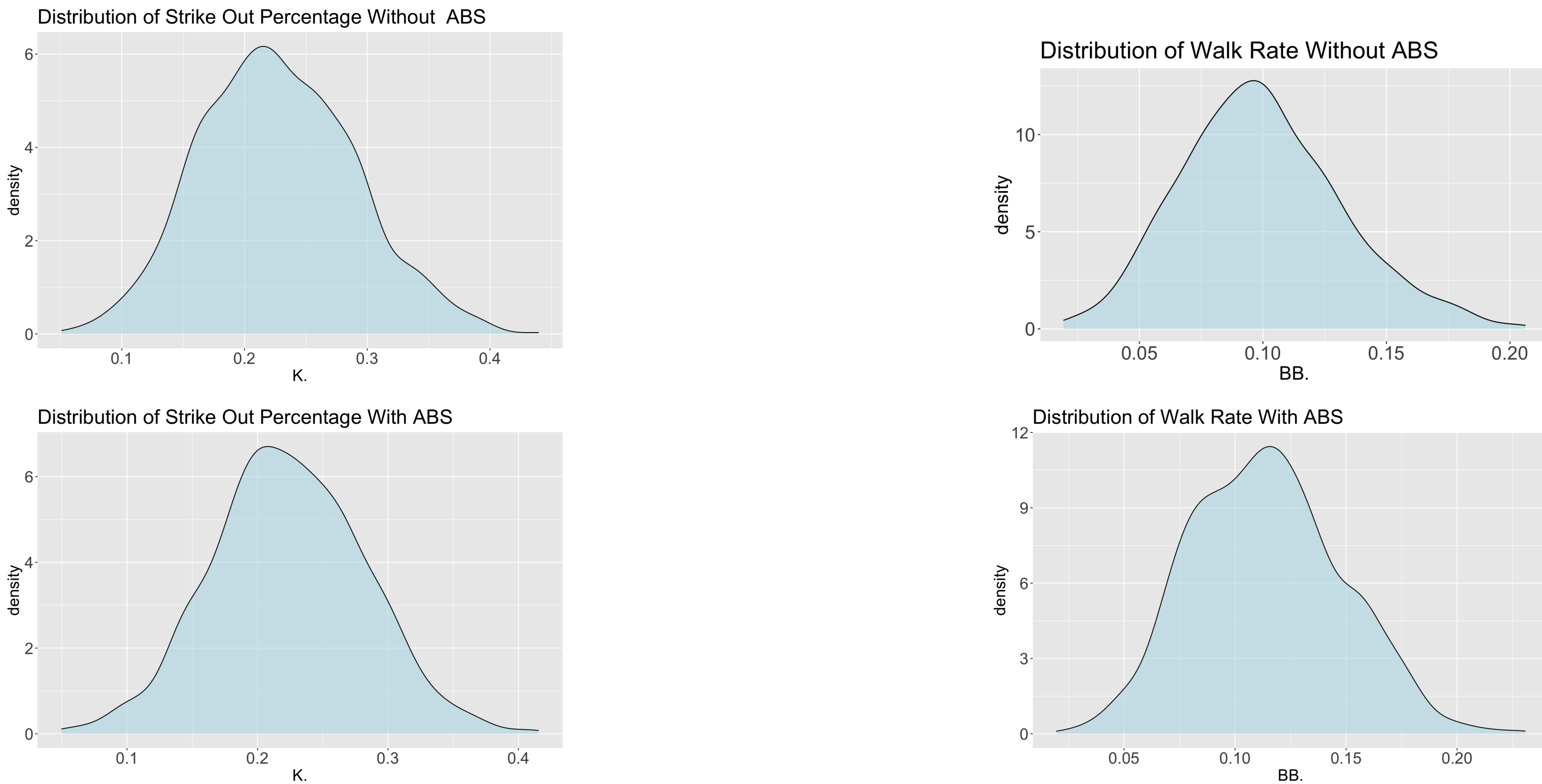
(b) **Pitches**

Figure 5: **Comparisons of annual vs aggregate statistics for Strikes and Pitches.**

From the graphs presented above, several clear trends emerge across the years 2019–2025. The most noticeable pattern is that nearly every metric shows a distinct change, either an increase or decrease, immediately following the introduction of the ABS system in 2023 (marked by the red vertical line). In the seasons that follow, most statistics begin to move back toward the overall league average across the entire period, represented by the dotted blue line. This convergence suggests that hitters in Triple-A may have gradually adapted to the new strike zone environment after its first full year of implementation. Observing these initial trends motivated a deeper investigation using more robust statistical tests to determine whether the changes in mean values across these metrics were significantly influenced by the introduction of the ABS system.

2.7 Hypothesis Tests and Density Plots

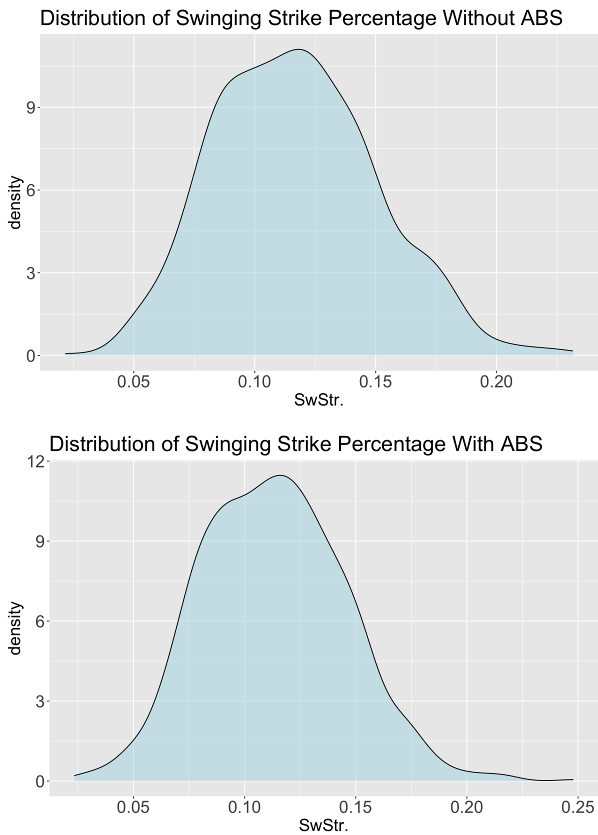
Before conducting any statistical tests, it was necessary to examine the distribution of each metric both before and after the implementation of the ABS system. To do this, the data were divided into two groups: Before ABS (2019–2022) and During ABS (2023–2025). This separation allowed for a direct comparison of performance metrics across the two periods. For each variable, two density plots were generated, one for each time frame, to assess whether the distributions were similar. This step ensured that the hypothesis tests would be based on valid and robust assumptions about the data.



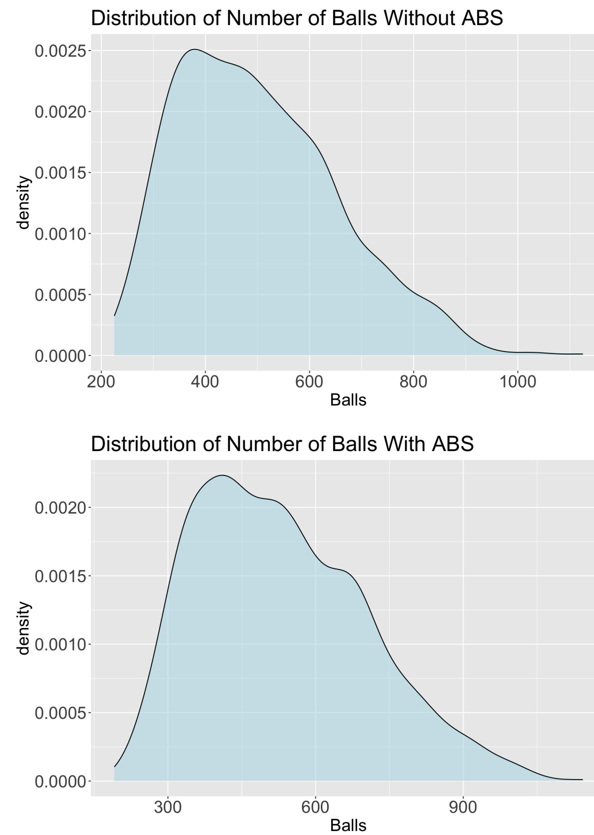
(a) Strikeout percentage (K%)

(b) Walk rate

Figure 6: Density plots for Strikeout percentage and Walk rate before and during ABS.

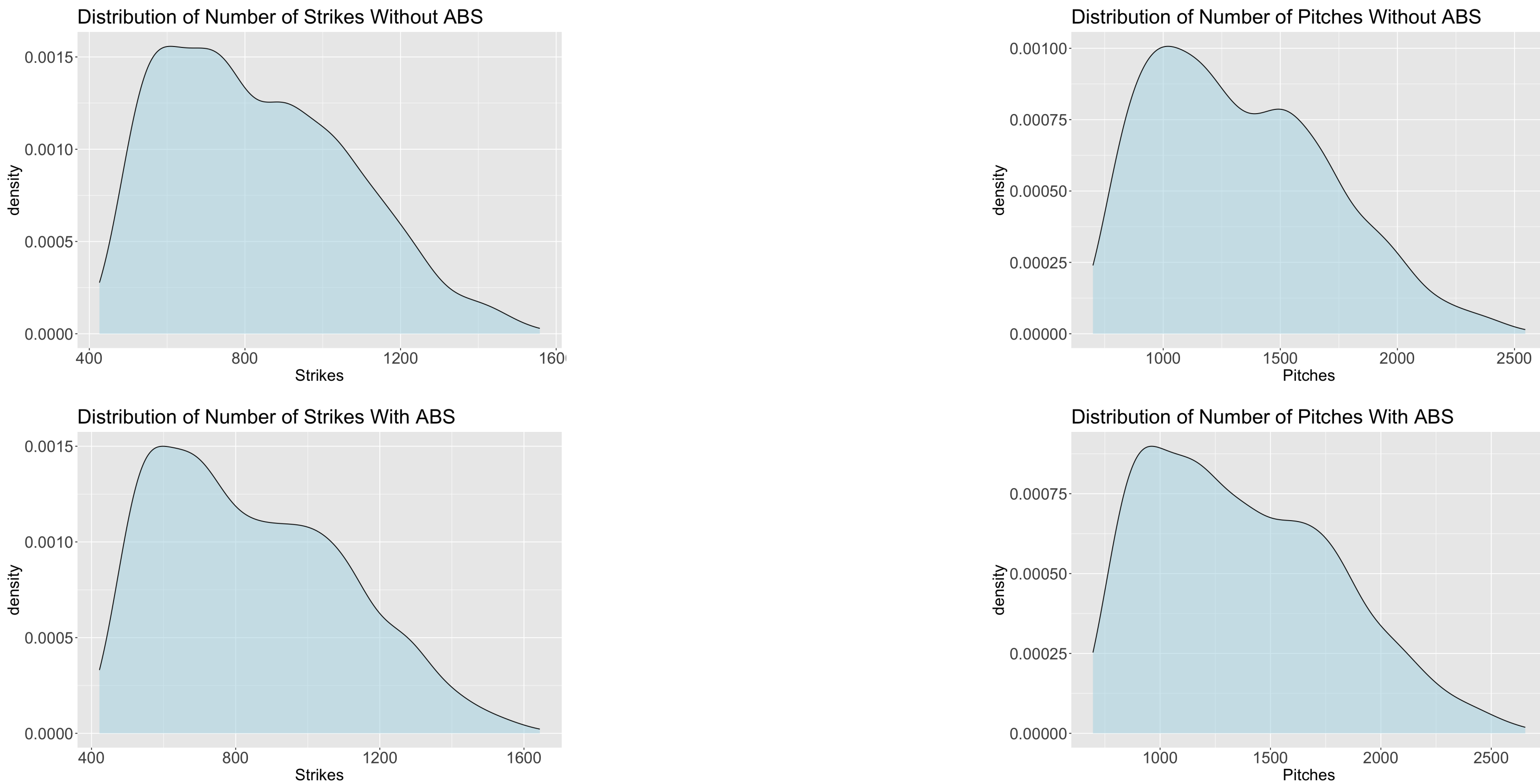


(a) Swinging Strike rate



(b) Balls

Figure 7: Density plots for Swinging Strike rate and Balls before and during ABS.



(a) Strikes

(b) Pitches

Figure 8: Density plots for Strikes and Pitches before and during ABS.

From these density plots, I was able to determine which hypothesis tests were most appropriate for evaluating whether the means of each metric changed significantly before and during the ABS period. For the metrics expressed as proportions, strikeout rate, walk rate, and swinging strike percentage, the distributions appeared approximately normal in both periods. Therefore, I did not hesitate to use a Welch's t-test to compare the means between the two groups, as this method does not assume equal variance across both groups.

For the other three metrics, number of pitches, balls, and strikes seen in a season, the distributions did not appear normal. However, given the large sample sizes for each group (982 observations before ABS and 1,086 during ABS), I applied the Central Limit Theorem to justify the use of the two sample z-test for these variables.

Given the large sample sizes, the t-test and z-test yielded nearly identical test statistics and confidence intervals, confirming that the choice of test did not affect the results.

Table 1: Comparison of Batting Metrics Before and During ABS

Batting Metric	**Mean (Before)**	**Mean (During)**	**Difference**	**Test Statistic**	**95% CI**
Strikeout percentage	0.2269	0.2238	0.0031	1.2085	$(-0.0019, 0.0084)$
Walk Rate	0.0993	0.1141	-0.0148	-10.186	$(-0.0177, -0.0120)$
Swinging Strike percentage	0.1170	0.1132	0.0038	2.6181	$(0.00095, 0.0066)$
Balls	500.86	530.06	-29.2	-4.0877	$(-43.21, -15.19)$
Strikes	833.82	849.68	-15.85	-1.4487	$(-37.32, 5.61)$
Pitches	1334.68	1379.74	-45.05	-2.5476	$(-79.73, -10.37)$

As shown in Table 1, four metrics exhibited statistically significant differences before ABS and after ABS periods: walk rate, swinging strike percentage, number of balls, and number of pitches.

- The walk rate in Triple-A increased by roughly one percentage point following the implementation of ABS, suggesting that hitters were drawing more walks during the automated strike zone time period.
- The swinging strike percentage was slightly higher before the introduction of ABS, indicating that hitters may have been better able to make contact under the new system.
- The number of balls and total pitches per season both rose after the implementation of ABS. This trend suggests that hitters may be experiencing longer at-bats and showing greater patience at the plate, possibly due to a tighter and more consistent strike zone. As a result, hitters could be seeing more favorable pitches, which may improve their chances of achieving positive outcomes at the plate.

Even though the results of the hypothesis tests don't have direct causality, seeing that the majority of the metrics exhibited significant changes between the pre-ABS and ABS periods, and that many of these means shifted immediately after implementation before gradually readjusting, motivated a deeper investigation into whether the impact of ABS varied across individual years. To explore this, I conducted a series of fixed effects regressions to estimate the year-by-year effects of ABS implementation on performance metrics in Triple-A Baseball.

2.8 Fixed Effects OLS Model

As discussed in the Mathematical Preliminaries, Fixed Effects Regression Models are well-suited for panel data, where observations are collected for the same subjects over multiple time periods. Given the structure of my dataset, tracking player-level metrics across seasons, a fixed effects framework was the appropriate choice to estimate the year-by-year effects of the ABS system in Triple-A Baseball. Importantly, the inclusion of player fixed effects allows the model to control for time-invariant, individual-specific characteristics (e.g., handedness, natural talent), therefore isolating the within-player variation over time. This helps provide a clearer view of the true impact that the ABS system may have had on hitter performance.

To capture these yearly effects, I included three indicator variables: abs2023, abs2024, and abs2025, representing the seasons in which the ABS system was implemented in Triple-A. Each indicator equals one if the player appeared in that specific year and zero otherwise. To construct these variables, I identified players who appeared in at least one pre-ABS season (2019–2022) and also participated in each respective ABS season. It is important to note that players could appear in multiple ABS years, but each player-year observation is treated as a distinct entity in the dataset. After applying this process, the dataset included 97 players with experience both before ABS and in 2023, 140 players in 2024, and 197 players in 2025.

In addition to these three indicator variables, I included age and number of plate appearances as independent variables in the regression. These variables help control for player experience and performance exposure, allowing for the possibility that older players may demonstrate greater plate discipline and that players with more plate appearances may exhibit improved patience at the plate.

Before estimating the model, I conducted a Variance Inflation Factor (VIF) test to assess potential multicollinearity among the independent variables. The results indicated that all variables were only minimally correlated (with VIF values just above 1), suggesting that multicollinearity was not a significant concern. Therefore, I proceeded with estimating the regression model by Ordinary Least Squares (OLS) without modification.

Fixed Effects OLS Models

Dependent Variables:	Strikeout Percentage	Walk Rate	Swinging Strike Percentage
Model:	(1)	(2)	(3)
Variables			
abs2023	-0.0106***	0.0162***	-0.0059***
	(0.0035)	(0.0024)	(0.0017)
abs2024	-0.0117**	0.0026	-0.0059**
	(0.0046)	(0.0031)	(0.0024)
abs2025	-0.0226***	5.6×10^{-5}	-0.0114***
	(0.0058)	(0.0039)	(0.0029)
Age	0.0044***	0.0017**	0.0012**
	(0.0013)	(0.0008)	(0.0006)
Plate Appearances	-9.04×10^{-6}	-1.21×10^{-5}*	-6.01×10^{-6}
	(1.07×10^{-5})	(7.21×10^{-6})	(4.95×10^{-6})
Fixed-effects			
PlayerId	Yes	Yes	Yes
Fit statistics			
Observations	1,500	1,500	1,500
R^2	0.80856	0.73876	0.84968
Within R^2	0.02040	0.11783	0.02666

Clustered (PlayerId) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Before interpreting the variables and their coefficients, it is important to clarify how statistical significance is represented in the regression table. Each coefficient is followed by asterisks to indicate its level of significance: three asterisks (***) denote significance at the 99% confidence level, two asterisks (**) indicate significance at the 95% confidence level, and one asterisk (*) represents significance at the 90% confidence level. The numbers shown in parentheses beneath each coefficient represent the clustered standard errors, which account for potential within-player correlation over time.

To begin interpreting the regression results, we first focus on the dependent variable strikeout percentage:

- For the 2023 ABS season, using the 97 players who appeared both before ABS and during 2023, the results indicate that players' strikeout percentages were 1.06 percentage points lower compared to their performance prior to ABS implementation.

- In 2024, based on the 140 players contributing to this indicator, the model shows a 1.17 percentage point decrease in strikeout percentage relative to their pre-ABS seasons, which is also statistically significant.

- In 2025, with a larger sample of 197 players, the model estimates a further 2.26 percentage point decrease in strikeout percentage compared to the pre-ABS period. This suggests that the effect of ABS on reducing strikeouts may have intensified over time.

- Lastly, the coefficient on Age is positive and statistically significant, implying that strikeout percentage tends to increase slightly as players get older. However, the magnitude of this effect is small, approximately a 0.44 percentage point increase per year of age, and therefore not practically meaningful.

Now, turning to the regression results for walk rate:

- Among the ABS indicators, the 2023 season shows the only significant increase in walk rate, a 1.62 percentage point rise for the 97 players in this sample. This finding suggests that early in ABS implementation, hitters may have become more patient or benefited from a tighter strike zone, while pitchers were still adjusting.

- Although the coefficients for Age and Plate Appearances are statistically significant, their magnitudes are small and thus not practically meaningful.

Finally, considering swinging strike percentage:

- For the 2023 and 2024 ABS seasons, the model estimates a nearly identical 0.59 percentage point decrease in swinging strike rate compared to the pre-ABS period. The small and consistent effect suggests that the impact of ABS may have stabilized after initial implementation, reflecting a plateau in how hitters adapted.
- In 2025, however, we observe a larger and statistically significant 1.14 percentage point decline in swinging strike rate among the 197 players in this sample. This may indicate a delayed adjustment period, where hitters continued to adapt and benefit from the tighter strike zone over time.
- Again, the coefficient for age is statistically significant but practically minor, representing only a 0.12 percentage point increase in swinging strike rate per year.

To assess the predictive strength of the models, we can examine the R^2 values for each dependent variable. Across all three models, the R^2 values range from approximately 0.74 to 0.85, indicating that the inclusion of player fixed effects allows the models to capture a substantial portion of the overall variation in the dependent variables.

The within-R^2 values, on the other hand, represent the proportion of variation within players over time that is explained by the independent variables only (the ABS indicator variables, age, and plate appearances). Although these values are relatively small, they suggest that much of the year-to-year variation in player performance is attributable to factors not captured by the model. Nevertheless, the consistently high levels of statistical significance for the ABS coefficients indicate that the system had a measurable and systematic effect on strikeout percentage, walk rate, and swinging strike percentage.

2.9 Fixed Effects Poisson Models

For the remaining three dependent variables, the number of balls, strikes, and pitches seen in a season, I employed a Fixed Effects Poisson Model to examine how these metrics changed over time. As outlined in the Mathematical Preliminaries, Fixed Effects Poisson Models are well-suited for static panel data with count-dependent variables. This modeling approach closely parallels the Fixed Effects regressions used in the previous section, with some key differences in how the coefficients and the plate appearances variable are interpreted.

Since the number of balls, strikes, and pitches naturally increases with the number of plate appearances, we cannot include plate appearances as an independent variable. Instead, a useful feature of Fixed Effects Poisson Models allows us to incorporate plate appearances as an offset term. The offset, entered as the logarithm of plate appearances, effectively controls for exposure by fixing its value in the regression, ensuring it does not directly influence coefficient estimates. This specification changes how we interpret the model coefficients. Rather than describing changes in total counts over a full season, the coefficients now reflect changes in the rate per plate appearance. Since the model uses a log link, each estimated coefficient (β) represents the log of the rate ratio. To express these effects in more intuitive terms, we can compute the expected percentage change per plate appearance using the transformation $e^{\beta} - 1$, where β is the coefficient on the variable being studied.

Fixed Effects Poisson Models

Dependent Variables:	Balls	Strikes	Pitches
Model:	(1)	(2)	(3)
Variables			
abs2023	0.0443***	-0.0161***	0.0070*
	(0.0075)	(0.0036)	(0.0038)
abs2024	-0.0004	-0.0067	-0.0044
	(0.0101)	(0.0046)	(0.0050)
abs2025	0.0011	-0.0116*	-0.0068
	(0.0129)	(0.0061)	(0.0066)
Age	0.0016	0.0004	0.0009
	(0.0028)	(0.0013)	(0.0014)
Fixed-effects			
PlayerId	Yes	Yes	Yes
Fit statistics			
Observations	1,500	1,500	1,500
Squared Correlation	0.97140	0.99249	0.99188
Pseudo R^2	0.83772	0.88880	0.92169
BIC	18,536.0	17,830.7	19,206.1

Clustered (PlayerId) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

For these interpretations, let us begin with the only variable that is not an indicator, Age. From our models, we can see that age of a hitter in this model is not a significant predictor in the expected percentage increase of number of balls, strikes, or pitches per plate appearance.

Turning to the first indicator variable, abs2023, which represents the 97 players who appeared both before and during the 2023 ABS season in Triple-A baseball:

Balls: The coefficient on abs2023 is 0.0443 and highly significant. Applying the exponential transformation $e^{\beta} - 1$, this corresponds to an expected 4.5% increase in the number of balls seen per plate appearance relative to pre-ABS seasons. To express this effect in more practical terms, we can multiply this rate increase by the typical number of plate appearances for players in this sample. For instance, with an average of 500 plate appearances per season, we would expect a batter to see approximately 23 more balls per season after the implementation of ABS.

Strikes: The coefficient on abs2023 for strikes is –0.0161, which is also statistically significant. Applying the same transformation, we find an expected 1.6% decrease in the number of strikes seen per plate appearance for the 97 player sample. Using the same 500-plate-appearance example, this translates to roughly 8 fewer strikes per season under the ABS system.

Pitches: The coefficient for abs2023 in the pitches model is significant at the 90% confidence level, but its magnitude is extremely small. After applying the exponential transformation, the implied percentage change is negligible, suggesting that the number of pitches per plate appearance remains essentially unchanged in 2023 compared to the pre-ABS period.

For the remaining two indicator variables, abs2024 and abs2025, the only statistically significant coefficient among the three models appears in the strikes model for 2025, which is significant at the 90% confidence level. This suggests a modest but measurable change in the rate of strikes seen per plate appearance during the 2025 ABS season. One possible interpretation is that, after the initial implementation period in 2023, pitchers may have begun to adjust their approach to the new, more consistent strike zone introduced by ABS. The diminished significance of other coefficients in 2024 and 2025 could indicate that the system and players gradually stabilized, reducing the magnitude of early-season disruptions observed in 2023.

For the fit statistics reported at the bottom the table, Squared Correlation and Pseudo R^2 the results can be interpreted as follows. The Squared Correlation and Pseudo R^2 values are quite high across all three models. This is not concerning, as these values are largely driven by the inclusion of player fixed effects, which absorb substantial between-player variation in outcomes. The relatively few significant explanatory variables suggest that most of the model's explanatory power comes from these fixed effects rather than from within-player variation. Note that Bayesian Information Criterion (BIC) does not have any relevance in interpreting the specific model results.

3 Results and Conclusions

From the two types of analyses, hypothesis tests and Fixed Effects regressions, we can draw several conclusions about the implementation of the ABS system in Triple-A baseball.

The hypothesis tests revealed statistically significant overall changes in walk rate, swinging strike rate, and the number of balls and pitches seen throughout the season. However, when examining the yearly Fixed Effects regression results, it becomes evident that the only year showing consistent and significant changes across these metrics was 2023. One possible explanation is that the introduction of such a complex and unfamiliar technology initially altered how both batters and pitchers approached each plate appearance. Although the estimated effects may appear numerically small, even minor percentage changes can have meaningful implications in the context of professional baseball.

At first glance, it may seem contradictory that the hypothesis tests and regression analyses tell slightly different stories about the impact of ABS. This difference arises from how the two methods treat the data. The hypothesis tests aggregate all seasons together to assess overall changes, while the Fixed Effects regressions isolate year-specific effects, allowing for a more detailed examination of when and to what extent ABS influenced player outcomes. This distinction underscores why the regression results provide a deeper understanding of the dynamics in the system's implementation.

When interpreting these findings, it is also essential to consider the sample sizes underlying each indicator variable in the regressions. Since the number of players included in each coefficient estimate is below 200 and multiple players contribute to several indicator variable estimates, there should be caution in attributing causality to the ABS system alone. Nonetheless, the consistent significance of the coefficients associated with the 2023 season suggests that the introduction of ABS likely had a measurable influence on player behavior and plate discipline. These findings highlight the need for continued research as Major League Baseball moves toward adopting ABS at the highest level, as similar adjustments and performance shifts may emerge among MLB players.

4 Further Research and Considerations

Due to Major League Baseball's upcoming implementation of the ABS Challenge System in 2026, I was eager to conduct this pilot study to gain a preview of how this technology has already impacted the game at the levels where it has been tested. However, because of data collection constraints, such as the limited technology available in Triple-A ballparks compared to Major League stadiums, the recent introduction of the system, and time limitations, the scope of this study was restricted to a surface-level analysis. Nonetheless, the future of baseball will undoubtedly provide more opportunities to collect richer data and broaden the scope of ABS-related research.

For example, with further data collection, a study like this could greatly benefit from employing a Difference-in-Differences (DiD) regression model. However, this type of model requires a sufficiently large and well-defined control group to make robust inferences. Since my dataset does not include leagues that have not seen ABS such as, Major League Baseball and Double-A, the present study was limited to examining changes within a single sample of Triple-A players. This study design then limits our ability to make causal inferences about the ABS system without controlling for more time-varying characteristics.

Another challenge in conducting studies at this level is the high degree of player mobility within the minor leagues. The average professional baseball player spends roughly three to five years in the minors, often moving frequently between Single-A, Double-A, and Triple-A teams [10]. This volatility makes it difficult to track players who accumulate a consistent number of plate appearances under the ABS system, which in turn can limit the ability to detect long-term effects. Once the system is implemented at the Major League level, it may allow for a more stable and comprehensive analysis, as players typically remain longer in the majors. Although the Challenge System will not apply to every pitch, the mere presence of ABS could still have meaningful psychological and behavioral effects on both pitchers and hitters.

The variables chosen for this study were selected based on baseball intuition, specifically, those metrics most interpretable and likely to be affected by ABS. However, future research could take advantage of the advanced technology in MLB stadiums to analyze more detailed data. Such data might reveal subtle but important changes in batter behavior that are not captured in this study's more traditional statistics. Additionally, incorporating detailed pitcher-level information (e.g., handedness, pitch type usage, or velocity profiles) would allow researchers to control for more confounding factors and isolate the true effect of ABS on batter outcomes.

For now, this pilot study provides early evidence suggesting that based on its three-year testing period in Triple-A baseball, the ABS system may influence batter discipline and plate appearance outcomes. While the observed effects are modest, they signal meaningful adjustments that could have larger implications once ABS becomes a permanent feature of the professional game.

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